higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 3

(6060463)

12 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only blue or black ink.
6. Write neatly and legibly.

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QUESTION 1

Technicians calibrate at the bench to ensure calibrations are effective and don't result in degradation of performance. They ensure that all components are in good working order prior to installation and can evaluate them when component failure is suspected.

1.1 Why is the bench recommended for calibration? (3)

1.2 Explain how you will go about performing calibration on an electronic pressure transmitter at the bench. (8)

1.3 Make a neat labelled drawing to show how you will calibrate the meter in QUESTION 1.2 showing ALL relevant connections. (12)

[23]

QUESTION 2

2.1 2.1.1 What is a thermistor? (2)

2.1.2 Make a neat labelled drawing of a thermistor. (6)

2.1.3 List FIVE applications of thermistors. (5)

2.1.4 How are thermistors constructed? (5)

2.1.5 Explain the working principle of a thermistor. (3)

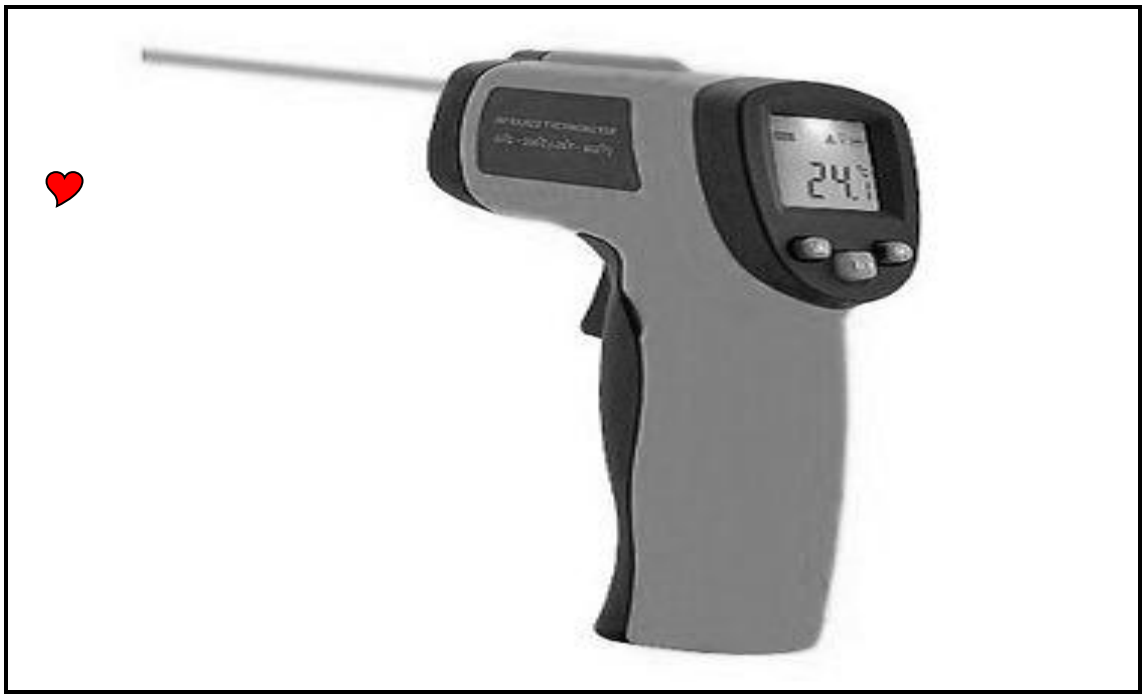
2.1.6 Complete the table below identifying the differences and similarities between thermistors and RTDs.

Write the answer next the letter (a–h) in the ANSWER BOOK.

RTDs	Thermistors
The materials used are copper, platinum, nickel, and tungsten	(a)
(b)	Have high hysteresis
Medium cost	(c)
(d)	Are small in size
Very small change in resistance	(e)
Here a conductor is used which gives an increase in resistance with an increase in temperature	(f)
(g)	The reproducibility is less
The range of measurements is -260 to 12 000 °C	(h)

(8 × 1) (8)

2.2



2.2.1 What instrument is shown in the picture? (1)

2.2.2 Calibration is essential in all devices. How will you go about calibrating the above instrument? (4)

2.2.3 State Stefan Boltzmann's law. (3)

[37]

QUESTION 3

3.1 Answer the following questions based an electromagnetic flowmeters.

3.1.1 Name FIVE limitations of electromagnetic flowmeters. (5)

3.1.2 Name FIVE advantages of electromagnetic flowmeters. (5)

3.1.3 State Faraday's law of magnetic induction. (3)

3.1.4 Make a neat labelled sketch of an electromagnetic flowmeter. (8)

3.1.5 Give THREE applications of electromagnetic flowmeters. (3)

3.2 List FIVE benefits of ultrasonic flowmeters. (5)

3.3 Explain how a vortex meter must be installed. (6)

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3.4 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.4.1–3.4.5) in the ANSWER BOOK.

3.4.1 Vortex flowmeters are well suited for measuring steam flow and they are widely used for this purpose.

3.4.2 A vortex flowmeter is a type of fluid meter that requires little or no maintenance.

3.4.3 The operating principle for the ultrasonic flowmeter requires a reliable high frequency sound transmitted across the pipe.

3.4.4 A magnetic flowmeter will generally work with hydrocarbons, distilled water and many non-aqueous solutions.

3.4.5 Magnetic flowmeters are not ideal for applications where low pressure drop and low maintenance are required.

(5 × 1) (5)

3.5 Choose the correct word or words from those given in brackets. Write only the answer next to the question number (3.5.1–3.5.5) in the ANSWER BOOK.

Vortex flowmeters offer many advantages for low measurements including 3.5.1 (easy/difficult) installation without impulse lines, 3.5.2 (no/with) moving parts to maintain or repair, 3.5.3 (less/more) leak potential and a 3.5.4 (wide/narrow) flow turndown range. They also offer very 3.5.5 (low/high) power consumption, allowing for use in remote areas.

(5 × 1) (5)

[45]

QUESTION 4

4.1

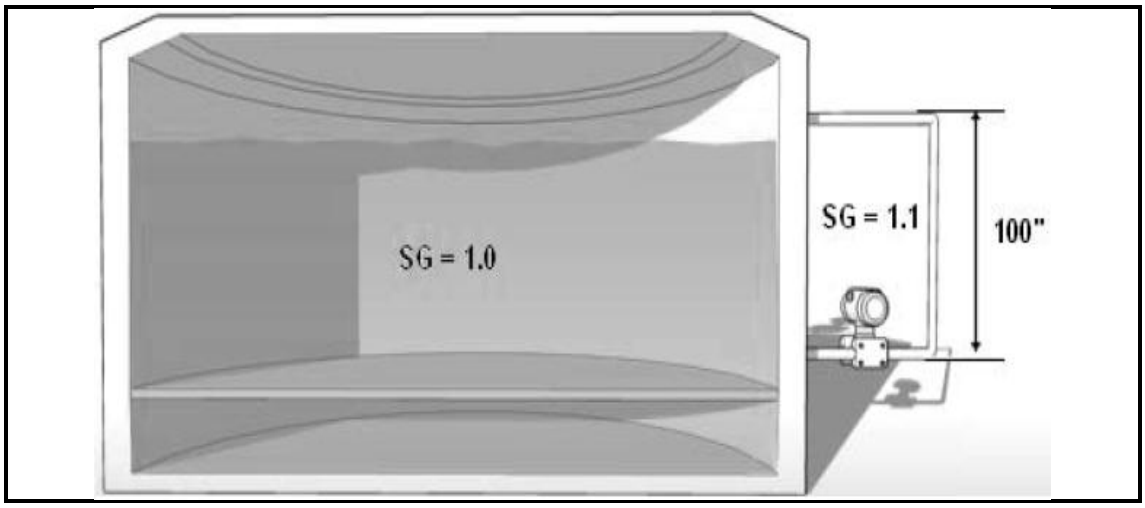


FIGURE 1

FIGURE 1 above represents a closed tank application where a 4 mA output is required when the level is at the bottom tap and a 20 mA output at the top tap. We also know that the low side leg is always filled with material whose specific gravity is 1.1. If the specific gravity of the fluid in the tank is 1.0, what is the calibration?

NOTE: Calibrate the transmitter to a range of-(-B) to (-B + S) (10)

4.2 Make drawings to show how the following meters are used to measure level.

4.2.1 Capacitive level meter (7)

4.2.2 Electromechanical level meter (7)

4.2.3 Conduction level meter (7)

4.3 What does suppressed-zero range mean? (4)

4.4 Level measurements.

4.4.1 Why is level measurement of importance in industries? (5)

4.4.2 Give FIVE different kinds of level measurement transmitters. (5)
[45]

TOTAL: 150

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